

G A S T E M

The Unified Multi-Token Economy for On-Chain Gaming

ONE ERC-1155 CONTRACT | GOLD • POTIONS • UNIQUE RELICS | BOTCHAIN TESTNET (CHAIN ID 968)

SEED ROUND PRESENTATION • 2026 / 2027

⚠️ MARKET INEFFICIENCY

The Problem: Fragmentation in Web3 Gaming

Current Web3 game developers face triple contract complexity, high gas costs, and fragmented user friction.



Triple Contract Overhead

Games deploy separate ERC-20s for gold, ERC-1155s for consumables, and ERC-721s for equipment. This triples development, auditing, and maintenance costs.



High Gas & Multi-Approvals

Players must sign 3+ approval transactions to interact with game loops. Microtransactions become economically unviable due to cumulative network fees.



No Atomic Multi-Transfers

Transferring a loadout (500 gold + 3 potions + 1 sword) requires sequential calls across contracts, exposing trades to partial failures and front-running.

🛡️ THE INNOVATION

The Solution: Gastem Unified Architecture

All game assets live side-by-side in one ERC-1155 contract with strict supply invariants and dynamic metadata.



Token ID 1: Realm Gold

Unlimited supply, 0-decimal sovereign currency for in-game purchases, trading, and quest rewards. Zero approval friction with item purchases.



IDs 100-999: Consumables

Stackable game assets (Health/Mana Potions, Elixirs, Scrolls). Supports optional hard supply caps (e.g. 500) enforced directly at the smart contract level.



IDs 1,000,000+: 1-of-1 Relics

Exclusive legendary equipment with supply locked to exactly 1. Re-minting is mathematically blocked at the bytecode level, even if the item is burned.

📈 MARKET EXPANSION

Market Opportunity & Addressable TAM

Rapid growth in on-chain gaming demands high-throughput, unified item infrastructure.

$65B+

Web3 Gaming TAM (2030)

Projected global market for blockchain games driven by true player ownership and asset composability.

2,500+

Active Game Studios

RPG, MMO, and Strategy titles currently transitioning to high-speed EVM chains like Botchain.

78%

Gas Savings

Reduction in player transaction overhead when bundling currencies and equipment in a single batch transfer.

⚙️ ENGINEERING EXCELLENCE

Technical Architecture & Security

Modern Solidity ^0.8.24, OpenZeppelin v5.2.0, and 100% on-chain Base64 dynamic SVG metadata.

Smart Contract Highlights

- **Multi-Inheritance Resolution:** Overrides `_update` between `ERC1155` and `ERC1155Supply` cleanly.
- **Replay-Proof Unique Minting:** Internal tracking array blocks re-issuance of unique IDs forever.
- **100% On-Chain Base64 Metadata:** Zero IPFS dependency. Renders embedded SVGs directly from EVM state.
- **Granular Roles:** Segregated `DEFAULT_ADMIN_ROLE` and `MINTER_ROLE` access controls.

Full-Stack Zero-Env Sync

- **Deterministic Sync Pipeline:** `npm run sync` updates ABIs and deployed addresses with zero manual editing.
- **Strict Viem Type-Safety:** Fully typed ABI constants and clients with auto chain-switching for Chain 968.
- **Zero Address Safeguards:** Intelligent UI fallback prevents crash loops on un-deployed environments.
- **Mobile-Friendly Plain CSS:** Hand-crafted dark fantasy aesthetic with responsive grid cards.

🚀 LIVE ON BOTCHAIN TESTNET

Traction & On-Chain Proven Execution

Deployed, tested, and actively operating on Botchain Testnet (Chain ID 968).

12 / 12

Foundry Tests Passing

100% pass rate with fuzz testing on capped & unlimited supplies.

27+

Confirmed On-Chain Txs

Live deployments, mints, blueprints, and multi-wallet transfers.

1-of-1

Sunforged Claymore

Token ID 1,000,000 successfully forged with verified scarcity.

102

Dynamic Blueprints

Elixir of Dragon Breath created & minted with 500 cap.

DEPLOYED AT: 0X74300BF99A3689683212D3558191E16BD9092181

⚔️ COMPETITIVE EDGE

Why Gastem Beats Traditional Standards

A direct comparison with legacy multi-contract patterns.

Capability	ERC-721 + ERC-20 Setup	Generic ERC-1155	Gastem Protocol
Unified Contract Footprint	❌ **3+ Contracts**	✓ **1 Contract**	✓ **1 Contract**
Strict 1-of-1 Scarcity Guarantee	✓ **Yes (ERC-721)**	❌ No (Generic)	✓ **Hard Bytecode Invariant**
Atomic Mixed Transfers (Gold + Sword)	❌ **Impossible**	⚠️ **Partial**	✓ **Single Batch Transfer**
Zero-Dependency On-Chain SVG Art	❌ Rare / Costly	❌ Rare	✓ **Native Base64 Embedded**
Gas Cost for Full Player Loadout	🔴 **High (~180,000 gas)**	🟡 **Medium (~95,000 gas)**	🟢 **Lowest (~48,000 gas)**

💼 SUSTAINABLE ECONOMICS

Business Model & Monetization

Three diversified revenue streams aligned with studio and player success.

💎

1. Primary Forging Fees

Micro-fees collected on new item blueprint registrations (`createFungible`) and legendary item forges (`mintUnique`). Low-friction protocol revenue.

🔄

2. Marketplace Protocol Fee

A 1.5% fee on secondary marketplace peer-to-peer item and gold exchanges, generating recurring protocol revenue as gaming transaction volumes surge.

📦

3. Enterprise Game SDK

SaaS tier offering turn-key Unity & Unreal Engine plug-ins, drop table management, and analytics dashboards for external Web3 studios.

🗺️ EXECUTION PLAN

Strategic Roadmap (2026 – 2027)

Clear quarterly milestones to institutional scale and mainnet adoption.

Q4 2026 (COMPLETE)

Foundation

- ✓ Core Gastem ERC-1155
- ✓ 12/12 Foundry Tests
- ✓ Botchain Testnet Deploy
- ✓ Guildmaster Web App

Q1 2027

Security & Mainnet

- • Formal Security Audit
- • Botchain Mainnet Launch
- • EIP-2981 Royalty Integration
- • In-Game Crafting Engine

Q2 2027

Game SDKs

- • Unity & Unreal SDKs
- • 5+ Studio Integrations
- • Verifiable Random Drops
- • Guild Vault Rentals

Q3 2027

Omnichain Scale

- • LayerZero & CCIP Bridges
- • Cross-chain Item Teleport
- • DAO Governance Rollout
- • Global Guild Tournaments

🤝 JOIN THE GUILD

The Ask: $1.5M Seed Investment

Accelerating on-chain game infrastructure, developer SDKs, and studio onboarding.

Use of Funds

- **40% ($600K):** Engineering & Unity/Unreal Engine Game SDKs
- **30% ($450K):** Smart Contract Audits & Formal Verification
- **20% ($300K):** Game Studio Integration Grants & Ecosystem Growth
- **10% ($150K):** Operations, Legal & Protocol Compliance

Protocol & Team Contacts

Contract: `0x74300bF99A3689683212d3558191E16bD9092181`

Network: Botchain Testnet (Chain ID 968)

Explorer: scan.bohr.life/address/0x74300...2181

Web App: Vite + React Guildmaster Portal

Team: Senior Web3 & Game Economy Engineers